

Principles Of Data Science

Numpy basics

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```
import numpy as np
import sys
S = range(1000)
print(sys.getsizeof(5)*len(S))
```

```
D = np.arange(1000)
print(D.size*D.itemsize)
```

```
28000
8000
```

```
v = np.ones((3,3))
print(v)
zeros_arr = np.zeros((2, 3))
print("\n", zeros_arr,"\n")
i_mat = np.eye(3)
print(i_mat)
```

```
[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]
```

```
[[0. 0. 0.]
 [0. 0. 0.]]
```

```
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

```
a = np.array([1,2,3,4])
b = np.array([4,3,2,4])
c = a>=b
print(c)
```

```
[False False  True  True]
```

```
a = np.array(42)
b = np.array([1, 2, 3, 4, 5])
c = np.array([[1, 2, 3],
              [4, 5, 6]])
d = np.array([
    [1, 2, 3],
    [4, 5, 6]],

    [[1, 2, 3],
     [4, 5, 6]]
```

```
)  
  
print(a.ndim)  
print(b.ndim)  
print(c.ndim)  
print(d.ndim)
```

```
0  
1  
2  
3
```

```
arr = np.arange(12)  
narr = arr.reshape((3,4),order='C')  
print(narr,"\n")  
narr = arr.reshape((3,4),order='F')  
print(narr)  
print(type(narr))
```

```
[[ 0  1  2  3]  
 [ 4  5  6  7]  
 [ 8  9 10 11]]
```

```
[[ 0  3  6  9]  
 [ 1  4  7 10]  
 [ 2  5  8 11]]  
<class 'numpy.ndarray'>
```

```
a = np.array([1., 2., 3., 4.])  
b = np.array([5., 6., 7., 8.])
```

```
print(a + b)  
print(a - b)  
print(a * b)  
print(a / b)  
print(a ** b)
```

```
[ 6.  8. 10. 12.]  
[-4. -4. -4. -4.]  
[ 5. 12. 21. 32.]  
[0.2          0.33333333 0.42857143 0.5          ]  
[1.0000e+00 6.4000e+01 2.1870e+03 6.5536e+04]
```

```
o=np.array([1,2,3,4])
```

```
f = o.flatten()
```

```
r = o.ravel()  
f[0] = 11  
print(f , r,o)
```

```
[11  2  3  4] [1 2 3 4] [1 2 3 4]
```

```
arr1 = np.array([[1, 2],  
                 [3, 4]])
```

```
arr2 = np.array([[5, 6],  
                 [7, 8]])
```

